Checkout — Request, Replay, Cancellation

Logical sequence and alternative outcomes | Review v1.4 | The matching .drawio file remains editable.

A. Customer -> API

POST /checkout; authenticated session + CSRF + Idempotency-Key

B. API -> Idempotency store

Find (customer_id, key); compare canonical explicit request

C. Existing COMPLETED

Same payload: replay original status/body; no cart lookup or mutation

D. Existing mismatch / active

Different payload: 409; concurrent IN_PROGRESS: retriable 409

E. New key

Reserve key; snapshot validated cart and shipping request

F. Mock payment

Simulate outcome; no real card or funds

G. Atomic DB transaction

Conditional stock decrement in deterministic product order

H. Atomic DB transaction

Persist order + immutable snapshots; clear cart; finalize result

I. Response

201 Created; repeat same key returns same stored 201/result

J. Failure / recovery

Rollback mutations on failure; interrupted processing requires safe recovery

K. Separate cancellation

CONFIRMED -> CANCELLED guarded transition + restore stock once